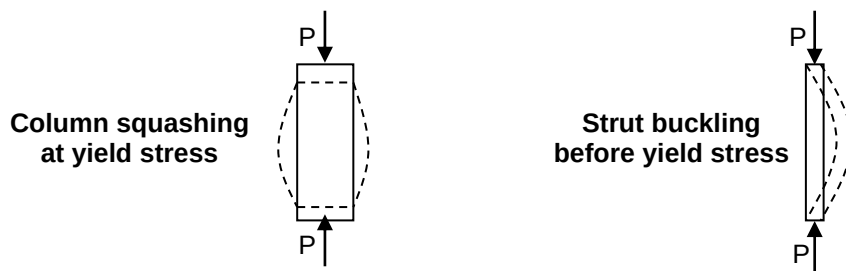


FCC3 232 CHAP 4 BUCKLING OF STRUTS

4.1 INTRODUCTION

Structural members which are predominantly subjected to **compression** can be classified into **two categories** depending on their **mode of failure**.

1. **Stocky columns** (referred to as **columns**) are compressive members whose failure mode is by **squashing when the yield stress is attained**.
2. **Slender columns** (referred to as **struts**) are compressive members whose failure mode is by **buckling before the yield stress is attained**.

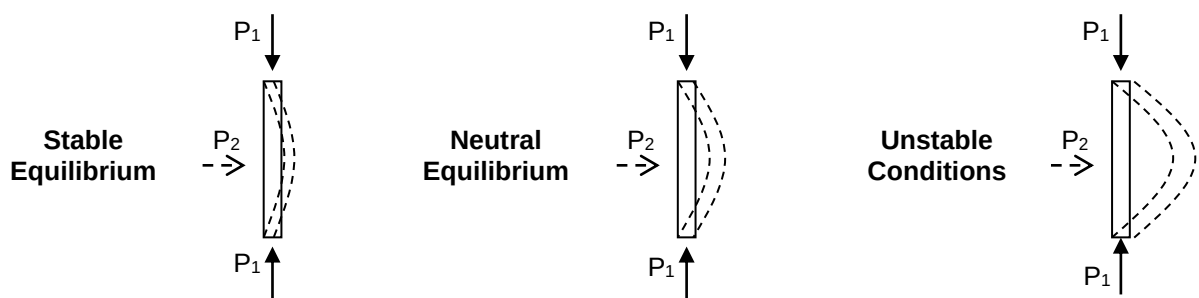


The analysis and design of **columns** is relatively easy because the yield stress for any material can be established in the laboratory by carrying out simple tests. This chapter presents the **theory of strut buckling** as pioneered by **Euler and Bernoulli (1757)**.

4.2 BUCKLING PHENOMENON

4.2.1 States of Equilibrium

Buckling is a form of **instability** which is exhibited by struts. Consider a strut which is loaded by the axial load P_1 , and which can be disturbed laterally by some small force P_2 .



When the load P_1 is **small**, if the strut is disturbed laterally by P_2 , it would spring back to the straight condition when the disturbing force is removed. Under these conditions, the strut is said to be in **stable equilibrium**. When P_1 attains the **critical value**, if the strut is disturbed laterally by P_2 , it would remain in the bend position, even after the disturbing force is removed. Under these conditions, the strut is said to be in **neutral equilibrium**. Any attempt to increase P_1 beyond the critical value would lead to **premature collapse by excessive bending**. Under these conditions, the strut is said to have become **unstable**.

4.2.2 Football Analogy

The concepts of stable equilibrium, neutral equilibrium and unstable equilibrium may be understood by using football analogy. Consider a ball which is placed on three different positions as illustrated in the diagram below.

Scenario 1: Ball on a trough: if the ball is displaced slightly, it will oscillate up and down, then settle at its original position.

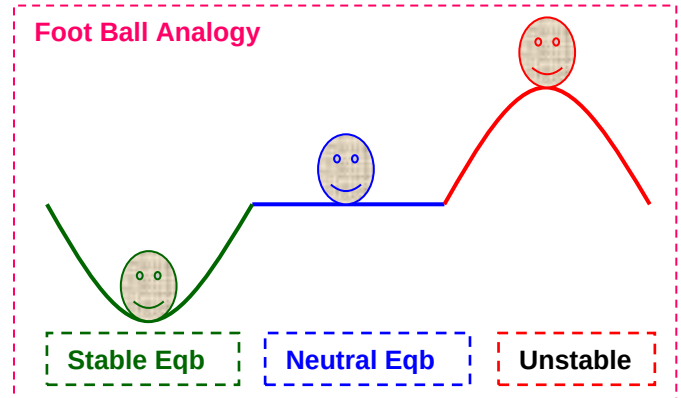
The ball is in **stable equilibrium**.

Scenario 2: Ball on a level surface

if the ball is displaced slightly, it will move slightly, then settle at its new position.

The ball is in **neutral equilibrium**.

Scenario 3: Ball on a crest: if the ball is displaced slightly, it will roll down the hill, it cannot stop itself, and the footballer has no control. The ball is said to be **unstable**.

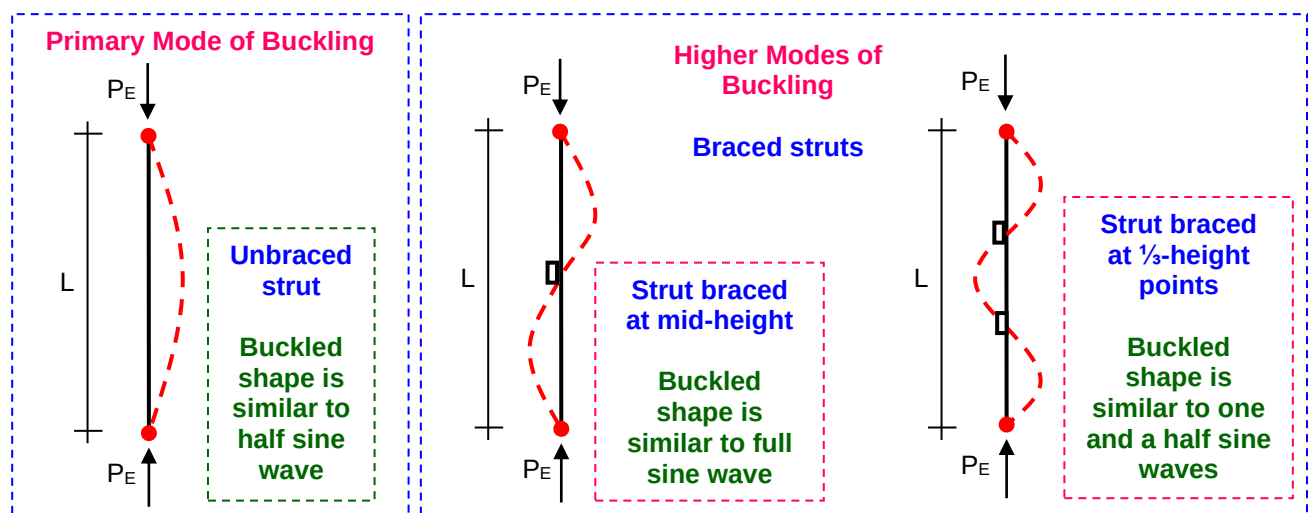


4.2.3 Buckling Modes

Consider a strut which is **not laterally supported** over its entire length. If the axial load is gradually increased up to the critical value, the strut will buckle by adopting a curved shape similar to a **half sine wave**. That shape is referred to as the **primary mode of buckling**.

Consider a strut which is **laterally supported at its mid-height**. If the axial load is gradually increased up to the critical value, the strut will buckle by adopting a curved shape similar to a **full sine wave**. That shape is referred to as a **higher mode of buckling**.

In structural design, the lateral supports are called **bracings**. If the strut is **laterally supported at one third height points**, the buckled shape will be similar to **one and a half sine waves**. The diagram below illustrates the primary mode and higher modes of buckling.



4.3 EULER'S THEORY OF STRUT BUCKLING [© 1757]

Euler's theory of strut buckling provides a criterion for predicting the **critical load** at which a strut will buckle. The following **assumptions** are made: -

1. The strut material is **homogeneous, isotropic** and **linearly elastic**.
2. The strut section is **prismatic** and **initially straight**.
3. The compressive **load is applied axially** along the **longitudinal axis** of the strut.
4. The strut **end restraints** can be idealized as either **pinned**, or **fixed** or **free**.

The **critical load** depends on the manner in which the strut is restrained (supported) at its ends. Euler investigated **four cases** which are outlined in the following **sub-sections**.

4.3.1 Case Study 1: Both Ends Pinned

Consider the strut shown below which has just buckled under the **Euler's critical load P_E** .

The coordinate system shown is used to find an expression

for the deflection equation using **Euler's beam theory**.

The bending moment at point C is given by $M_x = P_E y$

Using Euler's beam equation $-EI (d^2y/dx^2) = P_E y$

Rearrange $d^2y/dx^2 + (P_E/EI) y = 0$

If we let $P_E/EI = k^2$ then $d^2y/dx^2 + k^2 y = 0$

This is a standard 2nd order differential equation

whose solution is $y = C_1 \sin kx + C_2 \cos kx$

Apply boundary conditions at A and at B

At A, when $x = 0, y = 0 \Rightarrow C_2 = 0$

At B, when $x = L, y = 0 \Rightarrow C_1 \sin kL = 0$

If $C_1 \sin kL = 0 \Rightarrow$ Either $C_1 = 0$ OR $\sin kL = 0$

If $C_1 = 0 \Rightarrow y = 0$ for all $x \Rightarrow$ no buckling has occurred!

[this is called the **trivial solution** because the strut has just buckled].

If $\sin kL = 0 \Rightarrow kL = 0$ OR $kL = \pi$ OR $kL = 2\pi$ OR $kL = 3\pi$ etc

[Again, the solution $kL = 0$ is **trivial** because both k and L have real values for any strut]

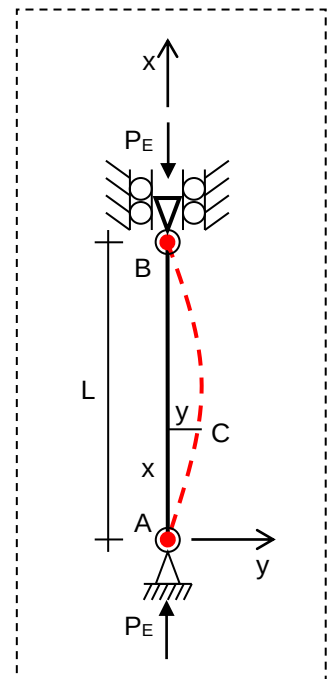
The least value of P_E to make the strut buckle is therefore given by **$kL = \pi$** .

[Higher values of kL represent higher modes of buckling]

If $kL = \pi \Rightarrow k^2 = P_E/EI = \pi^2/L^2 \Rightarrow P_E = \pi^2 EI/L^2$ ----- Eqn 1

The deflection equation becomes **$y = C_1 \sin (\pi/L) x$** which is a **sine wave**.

At mid-height $x = L/2 \Rightarrow y = C_1$. The constant C_1 is called the **amplitude** and cannot be determined from the above analysis.



4.3.2 Case Study 2: Both Ends Fixed

Consider the strut shown below which has just buckled under the **Euler's critical load P_E** . We need to find an expression for the deflection equation.

Let M be the fixing moments at the ends.

The bending moment at point C is given by $M_x = P_E y - M$

Using Euler's beam equation $-EI (d^2y/dx^2) = P_E y - M$

Rearrange $d^2y/dx^2 + (P_E/EI) y = (P_E/EI)(M/P_E)$

If we let $P_E/EI = k^2$ then $d^2y/dx^2 + k^2 y = k^2 M/P_E$

This is a standard 2nd order differential equation

whose solution is $y = C_1 \sin kx + C_2 \cos kx + M/P_E$

And $dy/dx = C_1 k \cos kx - C_2 k \sin kx$

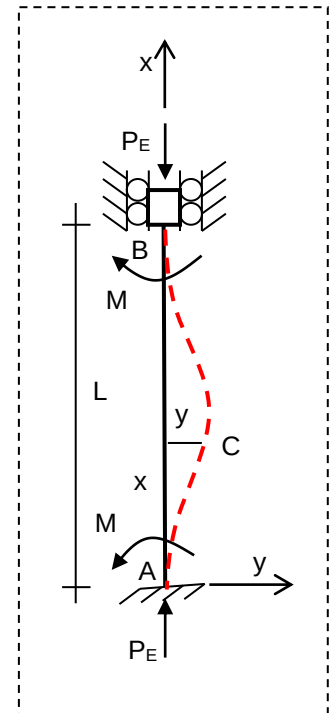
Apply boundary conditions at A

At A , when $x = 0, y = 0 \Rightarrow C_2 + M/P_E = 0 \Rightarrow C_2 = -M/P_E$

At A , when $x = 0, dy/dx = 0 \Rightarrow C_1 k = 0 \Rightarrow C_1 = 0$ since $k \neq 0$

The deflection equation becomes $y = -(M/P_E) \cos kx + M/P_E$

which can be expressed as $y = (M/P_E)(1 - \cos kx)$



Apply boundary condition at B

At B , when $x = L, y = 0 \Rightarrow 0 = (M/P_E)(1 - \cos kL) \Rightarrow \cos kL = 1$

If $\cos kL = 1 \Rightarrow kL = 0$ OR $kL = 2\pi$ OR $kL = 4\pi$ OR etc

[The solution $kL = 0$ is **trivial** since both k and L have real values for any strut]

The least value of P_E to make the strut buckle is therefore given by **$kL = 2\pi$** .

[Higher values of kL represent higher modes of buckling]

If $kL = 2\pi \Rightarrow k^2 = P_E/EI = 4\pi^2/L^2 \Rightarrow P_E = 4\pi^2 EI/L^2$ ----- Eqn 2

The deflection equation becomes $y = (M/P_E)(1 - \cos (2\pi/L)x)$ which is a **cosine wave**.

At mid-height $x = L/2 \Rightarrow y = 2M/P_E$. The **amplitude $2M/P_E$** cannot be determined from the above analysis because the fixing moment M is indeterminate.

4.3.3 Case Study 3: One End Fixed, the other End Pinned

Consider the strut shown below which has just buckled under the **Euler's critical load P_E** . There will be a horizontal reaction H , in order to maintain horizontal equilibrium.

We need to find an expression for the deflection equation.

Let M be the fixing moment at the fixed end A

The bending moment at point C is given by $M_x = P_E y - M + H x$

But $M = H L \Rightarrow H = M/L$

The bending moment at point C is given by $M_x = P_E y - M + (M/L) x$

Using Euler's beam equation $-EI (d^2y/dx^2) = P_E y - (M/L)(L - x)$

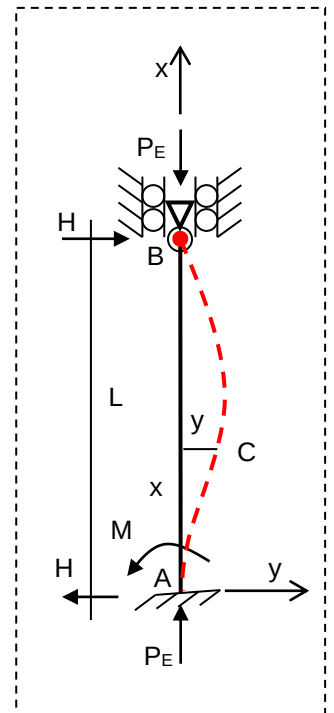
Rearrange $d^2y/dx^2 + (P_E/EI) y = (P_E/EI)(M/P_E L)(L - x)$

If we let $P_E/EI = k^2$ then $d^2y/dx^2 + k^2 y = k^2 (M/P_E L)(L - x)$

This is a standard 2nd order differential equation

whose solution is $y = C_1 \sin kx + C_2 \cos kx + (M/P_E L)(L - x)$

And $dy/dx = C_1 k \cos kx - C_2 k \sin kx - M/P_E L$



Apply boundary conditions at A

At A , when $x = 0, y = 0 \Rightarrow C_2 + M/P_E = 0 \Rightarrow C_2 = -M/P_E$

At A , when $x = 0, dy/dx = 0 \Rightarrow C_1 k - M/P_E L = 0 \Rightarrow C_1 = M/P_E k L$

The deflection equation becomes $y = (M/P_E k L) \sin kx - (M/P_E) \cos kx + (M/P_E L)(L - x)$

which can be expressed as $y = (M/P_E)[(\sin kx)/kL - \cos kx + 1 - x/L]$

Apply boundary condition at B

At B , when $x = L, y = 0 \Rightarrow 0 = (M/P_E)[(\sin kL)/kL - \cos kL] \Rightarrow \tan kL = kL$

The least value of P_E to make the strut buckle is approximated by $kL \approx \sqrt{2}\pi$.

If $kL \approx \sqrt{2}\pi \Rightarrow k^2 = P_E/EI \approx 2\pi^2/L^2 \Rightarrow P_E \approx 2\pi^2 EI/L^2$ ----- Eqn 3

NOTE: The **actual solution** to $\tan kL = kL$ is nearer $kL = 1.43029666\pi$, where kL is in **radians**. However, Euler's theory approximates this solution to $kL = \sqrt{2}\pi = 1.41421356\pi$. This approximation gives a **1% error** on the safe side.

4.3.4 Case Study 4: One End Fixed, the other End Free

Consider the strut shown below which has just buckled under the **Euler's critical load P_E** .

Let the lateral displacement at the free end be δ

We need to find an expression for the deflection equation.

The fixing moment at the fixed end A is given by $M = P_E \delta$

The bending moment at point C is given by $M_x = P_E y - M$

Using Euler's beam equation $-EI (d^2y/dx^2) = P_E y - P_E \delta$

Rearrange $d^2y/dx^2 + (P_E/EI) y = (P_E/EI) \delta$

If we let $P_E/EI = k^2$ then $d^2y/dx^2 + k^2 y = k^2 \delta$

This is a standard 2nd order differential equation

whose solution is $y = C_1 \sin kx + C_2 \cos kx + \delta$

And $dy/dx = C_1 k \cos kx - C_2 k \sin kx$

Apply boundary conditions at A

At A, when $x = 0$, $y = 0 \Rightarrow C_2 + \delta = 0 \Rightarrow C_2 = -\delta$

At A, when $x = 0$, $dy/dx = 0 \Rightarrow C_1 k = 0 \Rightarrow C_1 = 0$

The deflection equation becomes $y = 0 - \delta \cos kx + \delta$

which can be expressed as $y = \delta(1 - \cos kx)$

Apply boundary condition at B

At B, when $x = L$, $y = \delta \Rightarrow \delta = \delta(1 - \cos kL) \Rightarrow \cos kL = 0$

If $\cos kL = 0 \Rightarrow kL = \pi/2$ OR $kL = 3\pi/2$ OR $kL = 5\pi/2$ OR etc

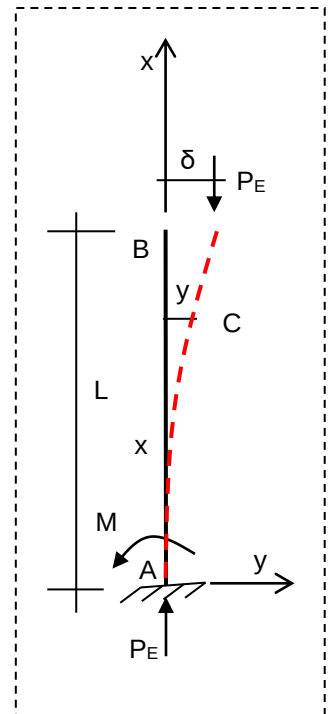
The least value of P_E to make the strut buckle is therefore given by **$kL = \pi/2$** .

[Higher values of kL represent higher modes of buckling]

If $kL = \pi/2 \Rightarrow k^2 = P_E/EI = \pi^2/4L^2 \Rightarrow P_E = \pi^2 EI/4L^2$ ----- Eqn 4

The deflection equation becomes $y = \delta(1 - \cos (\pi/2L)x)$ which is a **cosine wave**.

At the free end B, $x = L \Rightarrow y = \delta$. The **lateral displacement δ** cannot be determined from the above analysis.



4.4 BERNOULLI'S OBSERVATIONS [© 1757]

4.4.1 Effective Length

Bernoulli observed that **Euler's critical load** can be expressed in the **general format**:

$$P_E = \pi^2 EI / (\beta L)^2 \quad \text{-----} \quad \text{Eqn 5}$$

In case 1: Pinned end, Pinned end: $\beta = 1$ so that $P_E = \pi^2 EI / (L)^2 = \pi^2 EI / L^2$

In case 2: Fixed end, Fixed end: $\beta = 1/2$ so that $P_E = \pi^2 EI / (L/2)^2 = 4\pi^2 EI / L^2$

In case 3: Fixed end, Pinned end: $\beta \approx 1/\sqrt{2}$ so that $P_E \approx \pi^2 EI / (L/\sqrt{2})^2 \approx 2\pi^2 EI / L^2$

In case 4: Fixed end, Free end: $\beta = 2$ so that $P_E = \pi^2 EI / (2L)^2 = \pi^2 EI / 4L^2$

The coefficient β is called the **effective length factor**, and the quantity βL is called the **effective length** of the strut and is denoted by L_e . The effective length is defined by the

$$\text{equation} \quad L_e = \beta L \quad \text{-----} \quad \text{Eqn 6}$$

Equation 5 is therefore expressed as $P_E = \pi^2 EI / L_e^2 \quad \text{-----} \quad \text{Eqn 5*}$

4.4.2 Euler's Critical Stress

Bernoulli defined **Euler's critical stress** σ_E as the stress at which the strut buckles under **Euler's critical load** P_E given by Eqn 5*. If the strut has cross section area A , then Eqn 5* can be written as $P_E/A = \pi^2 E (I/A) / L_e^2$ and noting that the ratio $I/A = r^2$, then we can write $\sigma_E = \pi^2 E / (L_e/r)^2$. The ratio L_e/r is called the **slenderness ratio** of the strut and is denoted by λ . $\lambda = L_e/r = \beta L/r \quad \text{-----} \quad \text{Eqn 7}$

Where $r = \sqrt{I/A}$ is the **radius of gyration** about the axis of buckling (the minor axis)

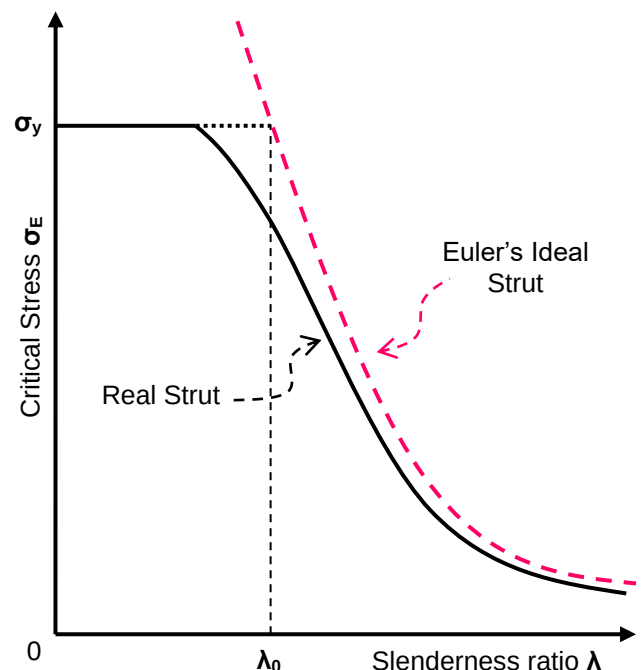
Euler's critical stress is therefore given by $\sigma_E = \pi^2 E / \lambda^2 \quad \text{-----} \quad \text{Eqn 8}$

4.4.3 Validity Limit for Euler's Theory

Bernoulli observed that a plot of σ_E against λ shows that **Euler's theory** is valid for struts with high λ . If the slenderness ratio is low, the strut becomes a column and fails by yielding. The **limiting slenderness ratio** which marks the boundary between columns and struts is denoted by λ_0 and can be evaluated using Eqn 8. $\sigma_y = \pi^2 E / \lambda_0^2$

$$\lambda_0 = \sqrt{(\pi^2 E / \sigma_y)} \quad \text{-----} \quad \text{Eqn 9}$$

The slenderness ratio λ_0 is called the **validity limit for Euler's theory**.



4.5 PERRY-ROBERTSON FORMULA

The behavior of a **real strut** deviates significantly from **Euler's ideal strut** because some of the assumptions made by Euler are difficult to achieve in practice. Examples are: -

1. In practice, it is difficult to apply the load axially. There will be some eccentricity.
2. In practice, it is difficult to get a straight strut. There will be some initial crookedness.

In addition, Euler's theory does not take account of the fact that buckling also depends on: -

1. The yield strength of the material. Different materials have different yield strengths.
2. The shape of the cross section. Different sections have different buckling strengths.

Research work carried out by **Perry et al (1886)** showed that the behavior of a real strut can be predicted by the equation:

$$(\sigma_y - \sigma_c)(\sigma_E - \sigma_c) = \eta \sigma_E \sigma_c \quad \text{-----} \quad \text{Eqn 10 (a)}$$

Where σ_y is the **yield stress** of the material; σ_c is the **buckling stress** of the strut
 σ_E is the **Euler's critical stress** of the strut; And η is called the **Perry factor**

Further research work carried out by **Robertson et al (1925)** showed that the **Perry factor** can be evaluated as:

$$\eta = 0.001 a (\lambda - \lambda_0) \geq 0 \quad \text{-----} \quad \text{Eqn 10 (b)}$$

Where λ is the **slenderness ratio** of the strut; λ_0 is the **limiting slenderness ratio**
 a is called the **Robertson constant** which depends on the shape of the section;
And **0.001** is the acceptable **tolerance** for both **crookedness** and load **eccentricity**.

In Eqn 10 (b), the proviso that $\eta \geq 0$ ensures that $\lambda \geq \lambda_0$ so that Eqn 10 (a) is only applicable to struts. When $\lambda < \lambda_0$ the **strut becomes a column**, and Eqn 10 (a) becomes inapplicable.

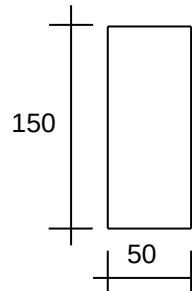
Eqns 10 (a) and 10 (b) are jointly referred to as the **Perry-Robertson Formula**, and they form the basis for the **design of struts**, as recommended in current codes of practice.

4.6 WORKED EXAMPLES

Example 1: Determine the **compressive capacity** and the **stress at failure** for a 1.8 m long timber compression member whose section is shown below if the end restraints are: -

- Both ends pinned
- Both ends fixed
- One end fixed, the other end pinned
- One end fixed, the other end free

[Take $E_t = 10 \text{ kN/mm}^2$ and $\sigma_y = 9 \text{ N/mm}^2$]



Solution [Compressive capacity is the load that a compression member can safely carry before its failure occurs]

$$[I_y = db^3/12 = 1562.5 \times 10^3 \text{ mm}^4 \quad r_y = \sqrt{I_y/A} = 14.4 \text{ mm} \quad \lambda_0 = \sqrt{(\pi^2 E / \sigma_y)} = 105]$$

- (a) $\lambda = \beta L/r = 1.0 \times 1800/14.4 = 125 > \lambda_0 \Rightarrow$ member will behave as a **strut**

$$P_E = \pi^2 EI/L^2 = 47.6 \text{ kN} \quad \sigma_E = \pi^2 E/\lambda^2 = 6.3 \text{ N/mm}^2$$

- (b) $\lambda = \beta L/r = (1/2) \times 1800/14.4 = 62.5 < \lambda_0 \Rightarrow$ member will behave as a **column**

$$P_y = A \sigma_y = 67.5 \text{ kN} \quad \sigma_y = 9.0 \text{ N/mm}^2$$

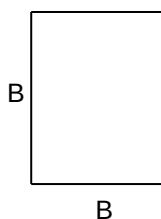
- (c) $\lambda = \beta L/r = (1/\sqrt{2}) \times 1800/14.4 = 88.4 < \lambda_0 \Rightarrow$ member will behave as a **column**

$$P_y = A \sigma_y = 67.5 \text{ kN} \quad \sigma_y = 9.0 \text{ N/mm}^2$$

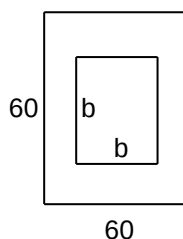
- (d) $\lambda = \beta L/r = 2.0 \times 1800/14.4 = 250 > \lambda_0 \Rightarrow$ member will behave as a **strut**

$$P_E = \pi^2 EI/4L^2 = 11.9 \text{ kN} \quad \sigma_E = \pi^2 E/\lambda^2 = 1.58 \text{ N/mm}^2$$

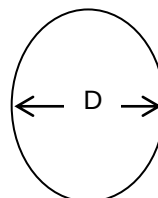
Example 2: Four compression members of length 1.2 m, whose both ends are pinned, and whose cross sections are shown below have equal cross section area $A = 2500 \text{ mm}^2$. The members are made of aluminium ($E = 70 \text{ kN/mm}^2$ and $\sigma_y = 450 \text{ N/mm}^2$). Determine the **slenderness ratio**, the **buckling load**, and the **stress at buckling** for each member. Comment on the relative strengths of the different types of cross sections.



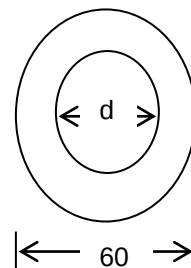
SECTION 1
Solid
Square
Section



SECTION 2
Hollow
Square
Section



SECTION 3
Solid
Circular
Section



SECTION 4
Hollow
Circular
Section

Solution $[\lambda_0 = \sqrt{(\pi^2 E / \sigma_y)} = 39.2]$

Section 1: $A = B^2 = 2500 \Rightarrow B = 50 \text{ mm}$ $I = B^4/12 = 520.8 \times 10^3 \text{ mm}^4$
 $r = \sqrt{(I/A)} = 14.4 \text{ mm}$ $\lambda = \beta L/r = 1.0 \times 1200/14.4 = 83.3 > \lambda_0 \Rightarrow \text{strut}$
 $P_E = \pi^2 EI/L^2 = 250 \text{ kN}$ $\sigma_E = \pi^2 E/\lambda^2 = 100 \text{ N/mm}^2$

Section 2: $A = (60^2 - b^2) = 2500 \Rightarrow b = 33.2 \text{ mm}$ $I = (60^4 - b^4)/12 = 978.8 \times 10^3 \text{ mm}^4$
 $r = \sqrt{(I/A)} = 19.8 \text{ mm}$ $\lambda = \beta L/r = 1.0 \times 1200/19.8 = 60.6 > \lambda_0 \Rightarrow \text{strut}$
 $P_E = \pi^2 EI/L^2 = 469.6 \text{ kN}$ $\sigma_E = \pi^2 E/\lambda^2 = 188.1 \text{ N/mm}^2$

Section 3: $A = \pi D^2/4 = 2500 \Rightarrow D = 56.6 \text{ mm}$ $I = \pi D^4/64 = 503.8 \times 10^3 \text{ mm}^4$
 $r = \sqrt{(I/A)} = 14.2 \text{ mm}$ $\lambda = \beta L/r = 1.0 \times 1200/14.2 = 84.5 > \lambda_0 \Rightarrow \text{strut}$
 $P_E = \pi^2 EI/L^2 = 241.7 \text{ kN}$ $\sigma_E = \pi^2 E/\lambda^2 = 96.8 \text{ N/mm}^2$

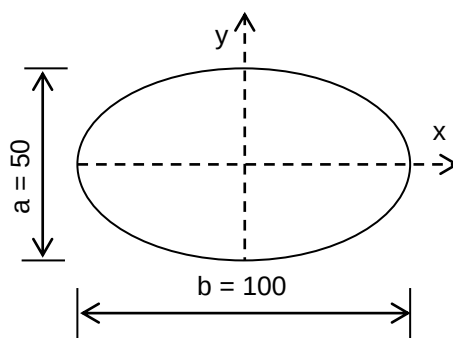
Section 4: $A = \pi(60^2 - d^2)/4 = 2500 \Rightarrow d = 20.4 \text{ mm}$; $I = \pi(60^4 - d^4)/64 = 627.7 \times 10^3 \text{ mm}^4$
 $r = \sqrt{(I/A)} = 15.8 \text{ mm}$ $\lambda = \beta L/r = 1.0 \times 1200/15.8 = 75.9 > \lambda_0 \Rightarrow \text{strut}$
 $P_E = \pi^2 EI/L^2 = 301.2 \text{ kN}$ $\sigma_E = \pi^2 E/\lambda^2 = 119.9 \text{ N/mm}^2$

Comment: Section 2 is strongest followed by 4, 1 and 3 in that order.

Example 3: A compression member of length 1.8 m, whose both ends are pinned, has an elliptical section with major diameter $b = 100 \text{ mm}$ and minor diameter $a = 50 \text{ mm}$ as shown below. The member is made of timber ($E = 10 \text{ kN/mm}^2$ and $\sigma_y = 9 \text{ N/mm}^2$).

Determine the **slenderness ratio**, the **buckling load**, and the **stress at buckling**.

[For an ellipse $A = \pi ab/4$ $I_x = \pi ba^3/64$ and $I_y = \pi ab^3/64$]



Solution $[\lambda_0 = \sqrt{(\pi^2 E / \sigma_y)} = 104.7]$

$A = \pi ab/4 = 3927 \text{ mm}^2$ $I_x = \pi ba^3/64 = 613.6 \times 10^3 \text{ mm}^4$ [buckling occurs about x-axis]
 $r = \sqrt{(I/A)} = 12.5 \text{ mm}$ $\lambda = \beta L/r = 1.0 \times 1800/12.5 = 144 > \lambda_0 \Rightarrow \text{strut}$
 $P_E = \pi^2 EI/L^2 = 18.7 \text{ kN}$ $\sigma_E = \pi^2 E/\lambda^2 = 4.8 \text{ N/mm}^2$